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Anhui University (SBIAHU)

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# Applied Calculus IV

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# Week 13

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**Recall** from calculus differentiation and integration are transforms

$$\frac{d}{dx}x^2 = 2x \quad \text{and} \quad \int x^2 dx = \frac{1}{3}x^3 + c.$$





# Recall

Linearity property: The transform of a linear combination of functions is a linear combination of the transforms. For  $\alpha$  and  $\beta$  constants

$$\frac{d}{dx} [\alpha f(x) + \beta g(x)] = \alpha f'(x) + \beta g'(x)$$

$$\int [\alpha f(x) + \beta g(x)] dx = \alpha \int f(x) dx + \beta \int g(x) dx$$

# Laplace Transform



## Definition (Laplace Transform )

Let  $f$  be a function defined for  $t \geq 0$ . The **Laplace transform** of  $f$ , denoted  $\mathcal{L}\{f(x)\}$ , is given by:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} \underbrace{e^{-st}}_{\text{red wavy line}} \underbrace{f(t) dt}_{\text{red underline}}$$

when the integral **converges**.

In the other words, the integral is said to be the Laplace transform of  $f$ , if the integral **converges**.

(the lim exists)

$$\int_0^{\infty} e^{-st} f(t) dt = \lim_{b \rightarrow \infty} \int_0^b e^{-st} f(t) dt$$

## Existence (Laplace Transform )

If

✓  $f$  is piecewise continuous on  $[0, \infty)$

✓  $f$  is of exponential order,

指数阶函数

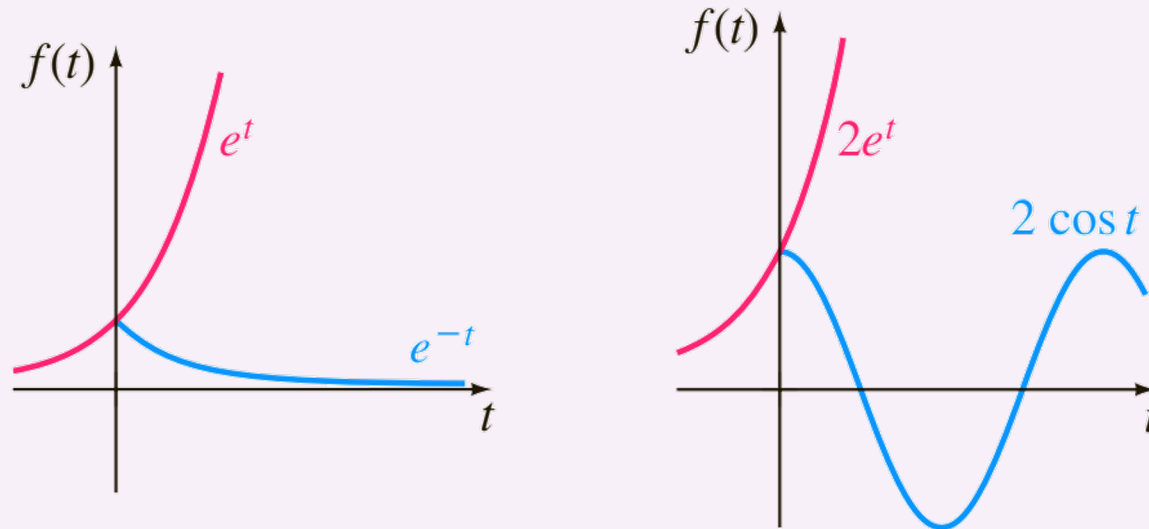
then  $\mathcal{L}\{f(t)\}$  exists for  $s > c$ .

### DEFINITION (Exponential Order)

A function  $f$  is said to be of **exponential order** if there exist constants  $c$ ,  $M > 0$ , and  $T > 0$  such that  $|f(t)| \leq Me^{ct}$  for all  $t > T$ .

## Examples

- ✓  $e^{-t}$  is of exponential order because we can introduce a function like  $Me^{ct}$  such that  $e^{-t}$  lies below  $Me^{ct}$  for all sufficiently large  $t$
- ✓  $2\cos t$  is of exponential order because we can introduce a function like  $Me^{ct}$  such that  $2\cos t$  lies below  $Me^{ct}$  for all sufficiently large  $t$





**Examples:** Evaluate  $\mathcal{L}\{1\}$



$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st}(1) dt$$

**Solution:**

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st}(1) dt = \lim_{b \rightarrow \infty} \int_0^b e^{-st} dt$$

$$= \lim_{b \rightarrow \infty} \left. \frac{-e^{-st}}{s} \right|_0^b$$

$$= \lim_{b \rightarrow \infty} \left( -\frac{e^{-sb}}{s} - \left( -\frac{e^{-s(0)}}{s} \right) \right)$$

$$= \lim_{b \rightarrow \infty} \left( -\frac{e^{-sb}}{s} + \frac{e^0}{s} \right)$$

$$= \lim_{b \rightarrow \infty} \left( -\frac{e^{-sb}}{s} + \frac{1}{s} \right) = -\frac{0}{s} + \frac{1}{s} = \frac{1}{s} \quad s > 0$$

$$\mathcal{L}\{1\} = \frac{1}{s} \quad \text{for } s > 0$$

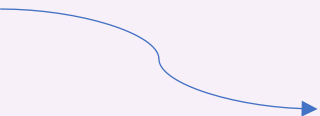
**Examples:** Evaluate  $\mathcal{L}\{t\}$



$$\mathcal{L}\{t\} = \int_0^{\infty} t e^{-st} dt$$

## Solution:

$$\mathcal{L}\{t\} = \int_0^{\infty} te^{-st} dt = \lim_{b \rightarrow \infty} \int_0^b te^{-st} dt$$


$$\int_0^b te^{-st} dt = \left[ -\frac{t}{s} e^{-st} \right]_0^b + \frac{1}{s} \int_0^b e^{-st} dt$$

$$= \lim_{b \rightarrow \infty} \left[ \left( -\frac{b}{s} e^{-sb} \right) - \left( -\frac{0}{s} e^{-s(0)} \right) \right] + \lim_{b \rightarrow \infty} \frac{1}{s} \left[ -\frac{1}{s} e^{-st} \right]_0^b$$

$$= \lim_{b \rightarrow \infty} \left[ -\frac{b}{s} e^{-sb} \right] + \lim_{b \rightarrow \infty} \left( -\frac{1}{s^2} e^{-sb} - \left( -\frac{1}{s^2} e^0 \right) \right)$$

$$s > 0 \quad = 0 + \frac{1}{s^2}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2} \quad \text{for } s > 0$$

**Examples:** Evaluate  $\mathcal{L}\{e^{-3t}\}$  and  $\mathcal{L}\{e^{5t}\}$

**Solution:**

$$\begin{aligned}\mathcal{L}\{e^{-3t}\} &= \int_0^{\infty} e^{-3t} e^{-st} dt = \int_0^{\infty} e^{-(s+3)t} dt \\ &= \left. \frac{-e^{-(s+3)t}}{s+3} \right|_0^{\infty} \\ &= \frac{1}{s+3}\end{aligned}$$

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3} \quad (s > -3)$$

**Solution:**

$$\begin{aligned}\mathcal{L}\{e^{5t}\} &= \int_0^{\infty} e^{5t} e^{-st} dt = \int_0^{\infty} e^{-(s-5)t} dt \\ &= \left. \frac{-e^{-(s-5)t}}{s-5} \right|_0^{\infty} \\ &= \frac{1}{s-5}\end{aligned}$$

$$\mathcal{L}\{e^{5t}\} = \frac{1}{s-5} \quad (s > 5)$$

**Examples:** Evaluate  $\mathcal{L}\{\sin 2t\}$



**Solution:**

$$\mathcal{L}\{\sin 2t\} = \int_0^{\infty} e^{-st} \sin 2t \, dt = \left. \frac{-e^{-st}}{s} \sin 2t \right|_0^{\infty} + \frac{2}{s} \int_0^{\infty} e^{-st} \cos 2t \, dt$$

$$= \frac{2}{s} \int_0^{\infty} e^{-st} \cos 2t \, dt, \quad s > 0$$

Integration by part

$$= \frac{2}{s} \left[ \left. \frac{-e^{-st} \cos 2t}{s} \right|_0^{\infty} - \frac{2}{s} \int_0^{\infty} e^{-st} \sin 2t \, dt \right]$$

$$s > 0 \qquad = \frac{2}{s^2} - \frac{4}{s^2} \mathcal{L}\{\sin 2t\}.$$

$$\text{After rearrangement} \rightarrow \mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}$$

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}, \quad s > 0$$

$\mathcal{L}$  is a linear transform

For  $\alpha$  and  $\beta$  constants

$$\int_0^{\infty} e^{-st} [\alpha f(t) + \beta g(t)] dt = \alpha \int_0^{\infty} e^{-st} f(t) dt + \beta \int_0^{\infty} e^{-st} g(t) dt$$

$$s > c$$

**Examples:** Evaluate  $\mathcal{L}\{3t + 5 \sin 2t + e^{-3t}\}$

$\mathcal{L}$  is a linear transform

$$\mathcal{L}\{3t + 5 \sin 2t + e^{-3t}\} = 3\mathcal{L}\{t\} + 5\mathcal{L}\{\sin 2t\} + \mathcal{L}\{e^{-3t}\}$$

**Solution:**

$$\mathcal{L}\{3t + 5 \sin 2t + e^{-3t}\} = 3\mathcal{L}\{t\} + 5\mathcal{L}\{\sin 2t\} + \mathcal{L}\{e^{-3t}\}$$


$$\mathcal{L}\{t\} = \frac{1}{s^2}$$
$$s > 0$$


$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3} \quad (s > -3)$$


$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}, \quad s > 0$$

$$\mathcal{L}\{3t + 5 \sin 2t + e^{-3t}\} = \frac{3}{s^2} + \frac{10}{s^2 + 4} + \frac{1}{s+3} \quad s > 0$$

So we can make a table of some basic functions and it can be perfectly used for finding the Laplace transform of many other functions

| Function $f(t)$ (for $t \geq 0$ )       | Laplace Transform         | Convergence Condition |
|-----------------------------------------|---------------------------|-----------------------|
| 1                                       | $\frac{1}{s}$             | $s > 0$               |
| $t$                                     | $\frac{1}{s^2}$           | $s > 0$               |
| $t^n$ ( $n = \text{positive integer}$ ) | $\frac{n!}{s^{n+1}}$      | $s > 0$               |
| $e^{at}$                                | $\frac{1}{s-a}$           | $s > a$               |
| $te^{at}$                               | $\frac{1}{(s-a)^2}$       | $s > a$               |
| $\sin(kt)$                              | $\frac{k}{s^2+k^2}$       | $s > 0$               |
| $\cos(kt)$                              | $\frac{s}{s^2+k^2}$       | $s > 0$               |
| $e^{at} \sin(kt)$                       | $\frac{k}{(s-a)^2+k^2}$   | $s > a$               |
| $e^{at} \cos(kt)$                       | $\frac{s-a}{(s-a)^2+k^2}$ | $s > a$               |

**Exercise:** show that

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

## Solution:

1. Start with the definition of  $\mathcal{L}\{t^n\}$ :

$$\mathcal{L}\{t^n\} = \int_0^{\infty} t^n e^{-st} dt = \lim_{b \rightarrow \infty} \int_0^b t^n e^{-st} dt$$

2. Apply Integration by Parts ( $\int u dv = uv - \int v du$ ):

- Let  $u = t^n \implies du = nt^{n-1} dt$
- Let  $dv = e^{-st} dt \implies v = -\frac{1}{s}e^{-st}$

$$\begin{aligned} \int_0^{\infty} t^n e^{-st} dt &= \left[ t^n \left( -\frac{1}{s} e^{-st} \right) \right]_0^{\infty} - \int_0^{\infty} \left( -\frac{1}{s} e^{-st} \right) nt^{n-1} dt \\ &\quad \downarrow \quad \quad \quad \downarrow \\ &\quad 0 \quad \quad \quad \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt \\ &\quad \quad \quad \downarrow \\ &\quad \quad \quad \frac{n}{s} \mathcal{L}\{t^{n-1}\} \end{aligned}$$

$$\Rightarrow \mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$



**Exercise:** show that

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

**Solution:**

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

we know that

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\} = \frac{n}{s} \left( \frac{n-1}{s} \mathcal{L}\{t^{n-2}\} \right)$$

$$= \frac{n \cdot (n-1)}{s^2} \mathcal{L}\{t^{n-2}\}$$

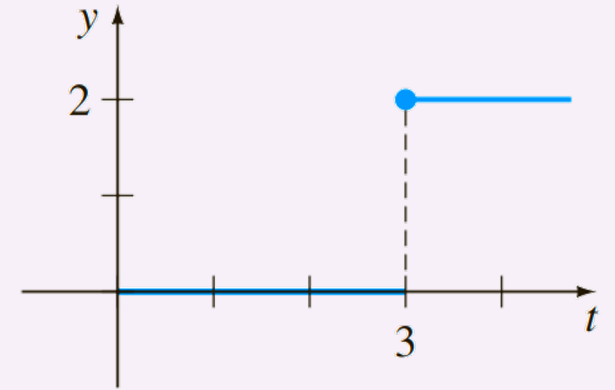
$$= \frac{n \cdot (n-1) \cdot (n-2)}{s^3} \mathcal{L}\{t^{n-3}\}$$

$\vdots$

$$= \frac{n \cdot (n-1) \cdot (n-2) \cdots 1}{s^n} \mathcal{L}\{t^0\} = \frac{n!}{s^n} \left( \frac{1}{s} \right)$$

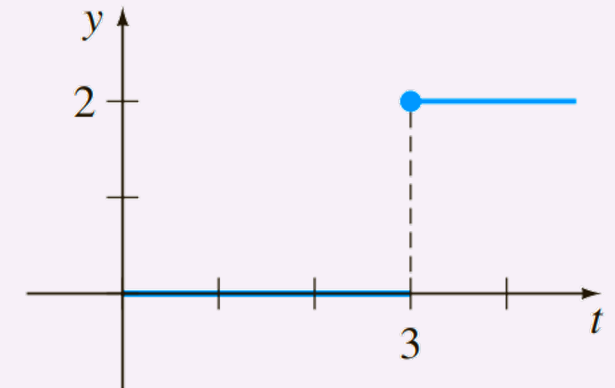
**Examples:** Evaluate  $\mathcal{L}\{f(t)\}$

$$f(t) = \begin{cases} 0, & 0 \leq t < 3 \\ 2, & t \geq 3. \end{cases}$$



**Solution:**

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st} f(t) dt = \int_0^3 e^{-st} (0) dt + \int_3^{\infty} e^{-st} (2) dt \\ &= 0 + \left. \frac{2e^{-st}}{-s} \right|_3^{\infty} \\ &= \frac{2e^{-3s}}{s}, \quad s > 0.\end{aligned}$$



## Exercises 7.1

In Problems 1–18 use Definition 7.1.1 to find  $\mathcal{L}\{f(t)\}$ .

$$1. f(t) = \begin{cases} -1, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

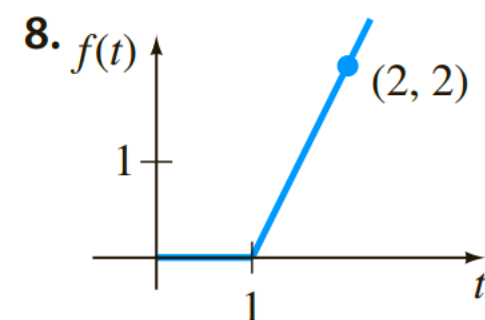
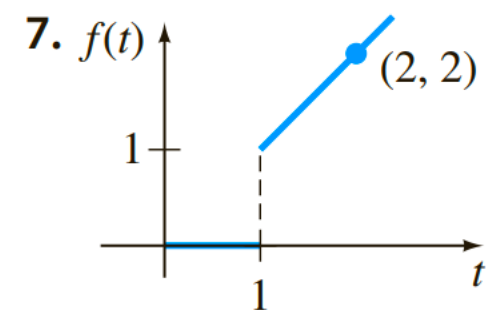
$$2. f(t) = \begin{cases} 4, & 0 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$

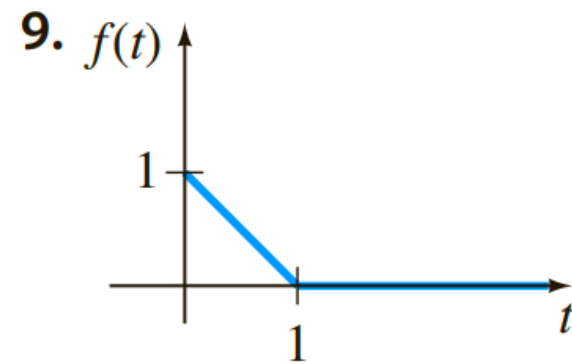
$$3. f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

$$4. f(t) = \begin{cases} 2t + 1, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$$

$$5. f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

$$6. f(t) = \begin{cases} 0, & 0 \leq t < \pi/2 \\ \cos t, & t \geq \pi/2 \end{cases}$$



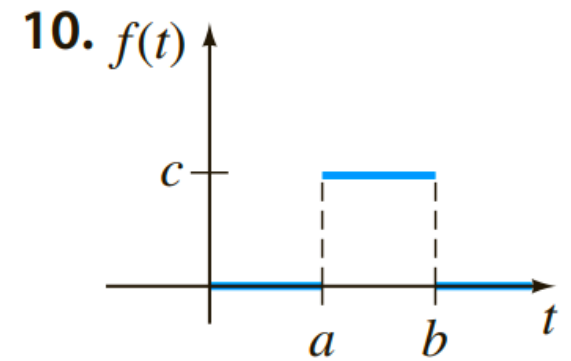


**11.**  $f(t) = e^{t+7}$

**13.**  $f(t) = te^{4t}$

**15.**  $f(t) = e^{-t} \sin t$

**17.**  $f(t) = t \cos t$



**12.**  $f(t) = e^{-2t-5}$

**14.**  $f(t) = t^2 e^{-2t}$

**16.**  $f(t) = e^t \cos t$

**18.**  $f(t) = t \sin t$

**19.**  $f(t) = 2t^4$

**21.**  $f(t) = 4t - 10$

**23.**  $f(t) = t^2 + 6t - 3$

**25.**  $f(t) = (t + 1)^3$

**27.**  $f(t) = 1 + e^{4t}$

**29.**  $f(t) = (1 + e^{2t})^2$

**31.**  $f(t) = 4t^2 - 5 \sin 3t$

**33.**  $f(t) = \sinh kt$

**35.**  $f(t) = e^t \sinh t$

**20.**  $f(t) = t^5$

**22.**  $f(t) = 7t + 3$

**24.**  $f(t) = -4t^2 + 16t + 9$

**26.**  $f(t) = (2t - 1)^3$

**28.**  $f(t) = t^2 - e^{-9t} + 5$

**30.**  $f(t) = (e^t - e^{-t})^2$

**32.**  $f(t) = \cos 5t + \sin 2t$

**34.**  $f(t) = \cosh kt$

**36.**  $f(t) = e^{-t} \cosh t$

In Problems 37–40 find  $\mathcal{L}\{f(t)\}$  by first using a trigonometric identity.

**37.**  $f(t) = \sin 2t \cos 2t$

**38.**  $f(t) = \cos^2 t$

**39.**  $f(t) = \sin(4t + 5)$

**40.**  $f(t) = 10 \cos\left(t - \frac{\pi}{6}\right)$



# Inverse Laplace Transform

If  $F(s)$  represents the Laplace transform of a function  $f(t)$ , that is

$$\mathcal{L}\{f(t)\} = F(s)$$

we then say  $f(t)$  is the **inverse Laplace transform** of  $F(s)$  and write:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

| Transform                                | Inverse Transform                                        |
|------------------------------------------|----------------------------------------------------------|
| $\mathcal{L}\{1\} = \frac{1}{s}$         | $1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$         |
| $\mathcal{L}\{t\} = \frac{1}{s^2}$       | $t = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$       |
| $\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$ | $e^{-3t} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$ |

**Examples:** Evaluate  $\mathcal{L}^{-1} \left\{ \frac{1}{s^5} \right\}$

## Solution:

We remember that  $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\} = \frac{1}{4!} \mathcal{L}^{-1}\left\{\frac{4!}{s^5}\right\} = \frac{1}{24} t^4.$$

**Examples:** Evaluate  $\mathcal{L}^{-1} \left\{ \frac{1}{s^2+7} \right\}$

## Solution:

We remember that  $\mathcal{L}\{\sin(kt)\} = \frac{k}{s^2 + k^2}$  (for  $s > 0$ )

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 7}\right\} = \frac{1}{\sqrt{7}} \mathcal{L}^{-1}\left\{\frac{\sqrt{7}}{s^2 + 7}\right\} = \frac{1}{\sqrt{7}} \sin\sqrt{7}t.$$

$\mathcal{L}^{-1}$  is a linear transform

For  $\alpha$  and  $\beta$  constants

$$\mathcal{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha \mathcal{L}^{-1}\{F(s)\} + \beta \mathcal{L}^{-1}\{G(s)\}$$

$$s > c$$



**Examples:** Evaluate  $\mathcal{L}^{-1} \left\{ \frac{-2s+6}{s^2+4} \right\}$

## Solution:

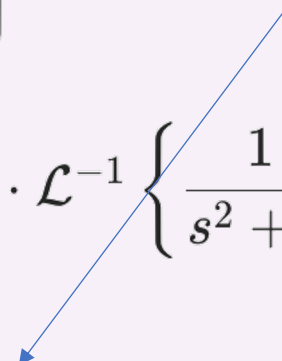
We know that

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} \right\} = \cos(2t)$$

$$\mathcal{L}^{-1} \left\{ \frac{2}{s^2 + 4} \right\} = \sin(2t)$$

$$\mathcal{L}^{-1} \left\{ \frac{-2s + 6}{s^2 + 4} \right\} = \mathcal{L}^{-1} \left\{ \frac{-2s}{s^2 + 4} + \frac{6}{s^2 + 4} \right\}$$

$$= -2 \cdot \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} \right\} + 6 \cdot \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 4} \right\}$$


$$6 \cdot \frac{1}{2} \cdot \mathcal{L}^{-1} \left\{ \frac{2}{s^2 + 2^2} \right\}$$

$$= -2 \cos(2t) + 3 \sin(2t)$$

**Examples:** Evaluate  $\mathcal{L}^{-1} \left\{ \frac{s^2 + 6s + 9}{(s-1)(s-2)(s+4)} \right\}$

**Solution:**

$$\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s + 4} \quad \Rightarrow \quad A = -\frac{16}{5}, B = \frac{25}{6}, \text{ and } C = \frac{1}{30}.$$

$$\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = -\frac{16/5}{s - 1} + \frac{25/6}{s - 2} + \frac{1/30}{s + 4},$$

$$\mathcal{L}^{-1} \left\{ \frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} \right\} = -\frac{16}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s - 1} \right\} + \frac{25}{6} \mathcal{L}^{-1} \left\{ \frac{1}{s - 2} \right\} + \frac{1}{30} \mathcal{L}^{-1} \left\{ \frac{1}{s + 4} \right\}$$

$$= -\frac{16}{5} e^t + \frac{25}{6} e^{2t} + \frac{1}{30} e^{-4t}$$

Note that

$$\mathcal{L}^{-1} \left\{ \frac{1}{s - a} \right\} = e^{at}$$

## Theorem

If  $f, f', \dots, f^{(n-1)}$  are continuous on  $[0, \infty)$  and are of exponential order and if  $f^{(n)}(t)$  is piecewise continuous on  $[0, \infty)$ , then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0),$$

where  $F(s) = \mathcal{L}\{f(t)\}$ .

**In particular,**

$$\mathcal{L} [f'] = s\mathcal{L} [f] - f(0)$$

$$\mathcal{L} [f''] = s^2\mathcal{L} [f] - sf(0) - f'(0)$$

$$\mathcal{L}[f'] = s\mathcal{L}[f] - f(0)$$

**Examples:** Use the Laplace transform to solve the initial-value problem

$$\frac{dy}{dt} + 3y = 13 \sin 2t, \quad y(0) = 6.$$

**Solution:** take the transform of each side of the differential equation:

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} + 3\mathcal{L}\{y\} = 13\mathcal{L}\{\sin 2t\}$$

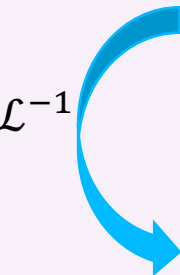
$$sY(s) - 6 + 3Y(s) = \frac{26}{s^2 + 4} \quad \Rightarrow \quad (s + 3)Y(s) = 6 + \frac{26}{s^2 + 4}$$

$$Y(s) = \frac{6}{s+3} + \frac{26}{(s+3)(s^2+4)} = \frac{6s^2+50}{(s+3)(s^2+4)}$$

$$\frac{6s^2+50}{(s+3)(s^2+4)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+4}$$

$A = 8, B = -2, C = 6$

$$Y(s) = \frac{6s^2+50}{(s+3)(s^2+4)} = \frac{8}{s+3} + \frac{-2s+6}{s^2+4}$$

$\mathcal{L}^{-1}$  

$$y(t) = 8\mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} + 3\mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\}$$

$$= 8e^{-3t} - 2\cos(2t) + 3\sin(2t)$$



**Examples:** Use the Laplace transform to solve the initial-value problem

$$y'' - 3y' + 2y = e^{-4t}, \quad y(0) = 1, \quad y'(0) = 5.$$

**Solution:**

$$\mathcal{L}[f''] = s^2 \mathcal{L}[f] - sf(0) - f'(0)$$

Take the transform of each side of the differential equation:


$$\mathcal{L}\{y'' - 3y' + 2y = e^{-4t}\}$$
$$\mathcal{L}\{y''\} - 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^{-4t}\}$$

$$s^2 Y(s) - sy(0) - y'(0) - 3[sY(s) - y(0)] + 2Y(s) = \frac{1}{s + 4}$$

$$(s^2 - 3s + 2)Y(s) = s + 2 + \frac{1}{s + 4}$$

$$Y(s) = \frac{s + 2}{s^2 - 3s + 2} + \frac{1}{(s^2 - 3s + 2)(s + 4)} = \frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)}$$

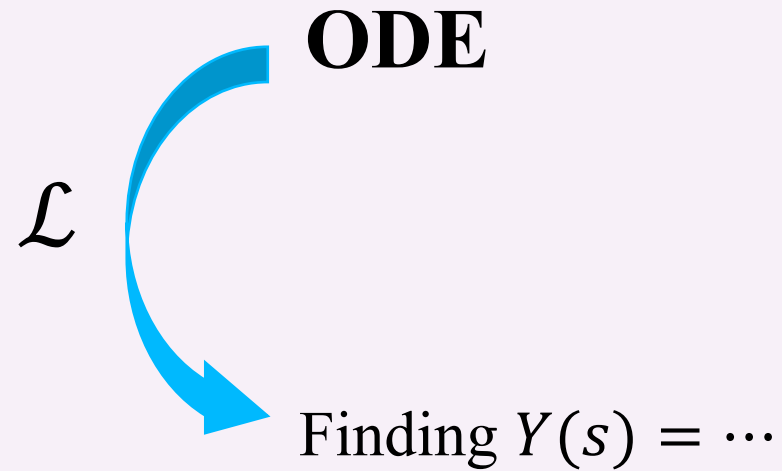
Decompse it

$$\frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s + 4} \quad \Rightarrow \quad A = -\frac{16}{5}, B = \frac{25}{6}, \text{ and } C = \frac{1}{30}$$

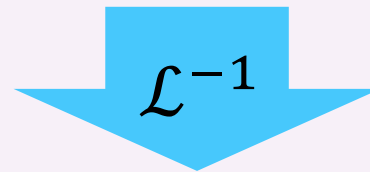
See the remark in page 292

$$\mathcal{L}^{-1} \curvearrowright \frac{s^2 + 6s + 9}{(s - 1)(s - 2)(s + 4)} = -\frac{16}{5(s - 1)} + \frac{25}{6(s - 2)} + \frac{1}{30(s + 4)}$$
$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = -\frac{16}{5}e^t + \frac{25}{6}e^{2t} + \frac{1}{30}e^{-4t}.$$

# Solving ODEs by Laplace transform method



Simplify, decompose or rearrange the relations,  
continue to get  $Y(s) = \dots$  in the easiest form for  $\mathcal{L}^{-1}$



$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

## Formulas for Laplace Transform of Derivatives

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}\{y'''(t)\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$$

$$\mathcal{L}\{y^{(4)}(t)\} = s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)$$

# Table of Laplace Transforms

|   | $f(t)$                         | $\mathcal{L}(f)$                |    | $f(t)$                 | $\mathcal{L}(f)$                      |
|---|--------------------------------|---------------------------------|----|------------------------|---------------------------------------|
| 1 | 1                              | $1/s$                           | 7  | $\cos \omega t$        | $\frac{s}{s^2 + \omega^2}$            |
| 2 | $t$                            | $1/s^2$                         | 8  | $\sin \omega t$        | $\frac{\omega}{s^2 + \omega^2}$       |
| 3 | $t^2$                          | $2!/s^3$                        | 9  | $\cosh at$             | $\frac{s}{s^2 - a^2}$                 |
| 4 | $t^n$<br>( $n = 0, 1, \dots$ ) | $\frac{n!}{s^{n+1}}$            | 10 | $\sinh at$             | $\frac{a}{s^2 - a^2}$                 |
| 5 | $t^a$<br>( $a$ positive)       | $\frac{\Gamma(a + 1)}{s^{a+1}}$ | 11 | $e^{at} \cos \omega t$ | $\frac{s - a}{(s - a)^2 + \omega^2}$  |
| 6 | $e^{at}$                       | $\frac{1}{s - a}$               | 12 | $e^{at} \sin \omega t$ | $\frac{\omega}{(s - a)^2 + \omega^2}$ |

|    | $F(s) = \mathcal{L}\{f(t)\}$                | $f(t)$                               |
|----|---------------------------------------------|--------------------------------------|
| 1  | $1/s$                                       | 1                                    |
| 2  | $1/s^2$                                     | $t$                                  |
| 3  | $1/s^n \quad (n = 1, 2, \dots)$             | $t^{n-1}/(n-1)!$                     |
| 4  | $1/\sqrt{s}$                                | $1/\sqrt{\pi t}$                     |
| 5  | $1/s^{3/2}$                                 | $2\sqrt{t/\pi}$                      |
| 6  | $1/s^a \quad (a > 0)$                       | $t^{a-1}/\Gamma(a)$                  |
| 7  | $\frac{1}{s-a}$                             | $e^{at}$                             |
| 8  | $\frac{1}{(s-a)^2}$                         | $te^{at}$                            |
| 9  | $\frac{1}{(s-a)^n} \quad (n = 1, 2, \dots)$ | $\frac{1}{(n-1)!} t^{n-1} e^{at}$    |
| 10 | $\frac{1}{(s-a)^k} \quad (k > 0)$           | $\frac{1}{\Gamma(k)} t^{k-1} e^{at}$ |
| 11 | $\frac{1}{(s-a)(s-b)} \quad (a \neq b)$     | $\frac{1}{a-b} (e^{at} - e^{bt})$    |
| 12 | $\frac{s}{(s-a)(s-b)} \quad (a \neq b)$     | $\frac{1}{a-b} (ae^{at} - be^{bt})$  |

## Table of Inverse Laplace Transforms

|    |                                      |                                                 |
|----|--------------------------------------|-------------------------------------------------|
| 13 | $\frac{1}{s^2 + \omega^2}$           | $\frac{1}{\omega} \sin \omega t$                |
| 14 | $\frac{s}{s^2 + \omega^2}$           | $\cos \omega t$                                 |
| 15 | $\frac{1}{s^2 - a^2}$                | $\frac{1}{a} \sinh at$                          |
| 16 | $\frac{s}{s^2 - a^2}$                | $\cosh at$                                      |
| 17 | $\frac{1}{(s - a)^2 + \omega^2}$     | $\frac{1}{\omega} e^{at} \sinh \omega t$        |
| 18 | $\frac{s - a}{(s - a)^2 + \omega^2}$ | $e^{at} \cos \omega t$                          |
| 19 | $\frac{1}{s(s^2 + \omega^2)}$        | $\frac{1}{\omega^2} (1 - \cos \omega t)$        |
| 20 | $\frac{1}{s^2(s^2 + \omega^2)}$      | $\frac{1}{\omega^3} (\omega t - \sin \omega t)$ |

## Table of Inverse Laplace Transforms



## Exercise:

Obtain the formulas given in the Tables of Laplace transform and the Table of Inverse Laplace transform in pages 58, 59 and 60

## Exercises 7.2

In problem 1-30, find the given inverse Laplace transform.

1.  $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$

3.  $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{s^5}\right\}$

5.  $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\}$

7.  $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s-2}\right\}$

2.  $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$

4.  $\mathcal{L}^{-1}\left\{\left(\frac{2}{s} - \frac{1}{s^3}\right)^2\right\}$

6.  $\mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3}\right\}$

8.  $\mathcal{L}^{-1}\left\{\frac{4}{s} + \frac{6}{s^5} - \frac{1}{s+8}\right\}$

$$9. \mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\}$$

$$11. \mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\}$$

$$13. \mathcal{L}^{-1}\left\{\frac{4s}{4s^2+1}\right\}$$

$$15. \mathcal{L}^{-1}\left\{\frac{2s-6}{s^2+9}\right\}$$

$$17. \mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\}$$

$$19. \mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\}$$

$$10. \mathcal{L}^{-1}\left\{\frac{1}{5s-2}\right\}$$

$$12. \mathcal{L}^{-1}\left\{\frac{10s}{s^2+16}\right\}$$

$$14. \mathcal{L}^{-1}\left\{\frac{1}{4s^2+1}\right\}$$

$$16. \mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2}\right\}$$

$$18. \mathcal{L}^{-1}\left\{\frac{s+1}{s^2-4s}\right\}$$

$$20. \mathcal{L}^{-1}\left\{\frac{1}{s^2+s-20}\right\}$$

$$21. \mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\}$$

$$22. \mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\}$$

$$23. \mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\}$$

$$24. \mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\}$$

$$25. \mathcal{L}^{-1}\left\{\frac{1}{s^3+5s}\right\}$$

$$27. \mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\}$$

$$29. \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\}$$

$$26. \mathcal{L}^{-1}\left\{\frac{s}{(s+2)(s^2+4)}\right\}$$

$$28. \mathcal{L}^{-1}\left\{\frac{1}{s^4-9}\right\}$$

$$30. \mathcal{L}^{-1}\left\{\frac{6s+3}{s^4+5s^2+4}\right\}$$

In Problems 35–44 use the Laplace transform to solve the given initial-value problem.

35.  $\frac{dy}{dt} - y = 1, \quad y(0) = 0$

36.  $2 \frac{dy}{dt} + y = 0, \quad y(0) = -3$

37.  $y' + 6y = e^{4t}, \quad y(0) = 2$

38.  $y' - y = 2 \cos 5t, \quad y(0) = 0$

39.  $y'' + 5y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0$

40.  $y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0) = 1, \quad y'(0) = -1$

41.  $y'' + y = \sqrt{2} \sin \sqrt{2}t, \quad y(0) = 10, \quad y'(0) = 0$

42.  $y'' + 9y = e^t, \quad y(0) = 0, \quad y'(0) = 0$

43.  $2y''' + 3y'' - 3y' - 2y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$

44.  $y''' + 2y'' - y' - 2y = \sin 3t, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$

Use the Laplace transform and to solve the given initial-value problem.

**45.**  $y' + y = e^{-3t} \cos 2t, \quad y(0) = 0$

**46.**  $y'' - 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 3$

**47.**  $y'' + 4y = 10 \cos 5t, \quad y(0) = 0, \quad y'(0) = 0$

**48.**  $y'' + 2y = 4t, \quad y(0) = 0, \quad y'(0) = 0$

$$21. \mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\}$$

$$22. \mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\}$$

$$23. \mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\}$$

$$24. \mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\}$$

$$25. \mathcal{L}^{-1}\left\{\frac{1}{s^3+5s}\right\}$$

$$27. \mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\}$$

$$29. \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\}$$

$$26. \mathcal{L}^{-1}\left\{\frac{s}{(s+2)(s^2+4)}\right\}$$

$$28. \mathcal{L}^{-1}\left\{\frac{1}{s^4-9}\right\}$$

$$30. \mathcal{L}^{-1}\left\{\frac{6s+3}{s^4+5s^2+4}\right\}$$